SILAJEET BANERJEE

MSc in Computer Science, RKMVERI, Belur

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GitHub
LinkedIn
Portfolio

SUMMARY

Computer Science graduate with interests in Artificial Intelligence, Machine Learning, and intelligent systems. Skilled in software development, machine learning, and intelligent systems with experience building AI-driven applications and autonomous platforms. Seeking opportunities to apply analytical and research skills in Artificial Intelligence, Data Science, and intelligent systems.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Computer Science, 2025–27
CGPA – 7.73(till Sem 2)

University of Calcutta, Kolkata
BSc (Honours) in Computer Science, 2021–24
CGPA –7.039

SKILLS

- **Languages:** C, C++, Java, Python, SQL
- **Frameworks & Libraries:** PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, Streamlit, Matplotlib, LangChain, LangGraph, AutoGen, crewAI
- **Operating Systems:** Windows, Linux
- **Tools:** Git, LaTeX, Jupyter Notebooks

Research Experience

ACM IKDD Uplink Summer Research Intern
May 2026 – Present
IIT Bombay, Mumbai, India
Supervisor: **Prof. Harshad Khadilkar**

- Designing and implementing a scalable, privacy-preserving, modular agentic pipeline for extracting and identifying research papers from Core A* venues with Indian-affiliated authors.

COURSE WORK

- Machine Learning, Deep Learning, Artificial Intelligence
- Probability, Linear Algebra, Mining of Massive Datasets
- Design and Analysis of Algorithms, Advanced Algorithms

ACHIEVEMENTS

- CUET-PG 2025 (Computer Science) – Qualified
- Awarded an Honorable Mention for the technical depth and compelling communication of my project work in the IKDD Project Presentations. Certificate
- Presented my project "Beyond the Hub: Exploiting Structural Bridges for Targeted Epidemic Control via Graph Reinforcement Learning" at the Ajit Diwan Memorial Workshop (2026).

PROJECTS

SENTRY: A Structured Extraction, Normalization, and Tiered Retrieval System for Scholarly Affiliation Tracking

IIT Bombay May 2026 – Present

Architected SENTRY, a 4-tier local web-agent replacing reliance on generic web agents such as Agent-E used in earlier systems for maintaining IKDD Premier Papers portal, cutting required LLM scale from 70B to 20B parameters with 100% Apple Silicon execution and zero API costs. Processed 36,595 papers across 26 venues at a 0.082% error rate using bidirectional rule-LLM filters to resolve acronym collisions. Integrated OpenWebUI and Selenium RPA with Jaccard deduplication for automated IKDD portal nominations.

A.P.R.I.L.

Independent Open Source Project May 2026 – Present

Engineered a local cognitive voice assistant using Whisper STT, Kokoro TTS, and local LLMs for real-time speech interaction. Architected a dual-stage memory substrate combining a sliding-window working context with a local Neo4j graph database for long-term semantic persistence. Built an MCP server ecosystem with deterministic system guardrails to decouple tool routing, enabling multi-hop entity resolution without conversational drift.

NetGuard AI – Rumor Containment via Graph Reinforcement Learning

RKMVERI Sep 2025 – Dec 2025

Engineered a custom environment to simulate stochastic information diffusion (SIR dynamics) on modular social networks (Caveman graphs). Developed and compared MCTS, DQN, and GraphSAGE agents, designed an aggressive reward shaping mechanism to penalize total infection duration. Selected for presentation at the Ajit Diwan Memorial Workshop (January 2026), recognized for novel application of GNNs in viral containment.

StockLLM

RKMVERI Feb 2026 – Apr 2026

Developed an AI-driven stock market forecasting system combining iTransformer and Temporal Fusion Transformer (TFT) models with technical indicators and live news sentiment analysis. Generated 5-day price forecasts and consensus BUY/SELL/HOLD recommendations using an LLM-powered decision orchestration pipeline.

NeuroGraph-X – Hybrid CNN-GNN for Alzheimer's Detection

RKMVERI Feb 2026 – Apr 2026

Developed NeuroGraph-X, a hybrid CNN-GNN framework for early Alzheimer's progression prediction (pMCI vs sMCI) using structural MRI data from the ADNI dataset. Designed dual spatial and morphological brain graphs with edge-aware GATv2 attention and cross-attention fusion, achieving 73% recall for progressive MCI detection.

AI-Assisted Autonomous Ground Vehicle (AGV)

University of Calcutta Apr 2024 – Jul 2024

Developed an AI-assisted autonomous robotic platform integrating YOLOv5-based human detection, real-time target tracking, and wireless control using ESP32, ESP8266, and Arduino microcontrollers. Implemented computer vision-guided servo actuation and remote monitoring for autonomous surveillance and target localization.